AID843 Assignment 3 Report

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***Abstract*—As a part of the course AID843 - Spatio-temporal Data Analytics, this report contains the results of and challenges associated with the completion of tasks 1-6 of Assignment 3, covering spatiotemporal processing, feature engineering, ML model training and data mining of the NHL Clean Shots (2020-2021 to 2024-2025) dataset.**

I. Dataset Description

The dataset is the NHL clean shots dataset, sourced from publicly available play-by-play tracking data. Each record corresponds to a single shot event and contains raw coordinates, the attending team identifier, the home team's defending side, period number, clock time remaining in the period, shot type, and the season.

A critical preprocessing step normalises all coordinates so that every attacking team shoots from left to right, placing the target goal at $(89, 0)$ in feet. This adjustment is determined per-event from `eventOwnerTeamId`, `homeTeamId`, and `homeTeamDefendingSide`, producing `x_adj` and `y_adj` fields used uniformly across all downstream analysis. After filtering to the three relevant event types and dropping records with missing coordinates, the dataset spans tens of thousands of games and hundreds of thousands of shot attempts across multiple NHL seasons. The target label `is_goal` is a binary indicator derived from `eventType`.

II. Feature Engineering

A. Spatial Features

Spatial features are derived entirely from the adjusted coordinate pair (`x_adj`, `y_adj`) relative to the fixed goal position $(89, 0)$.

Euclidean shot distance d and shot angle θ are computed and Zone membership partitions the offensive half into eight mutually exclusive Boolean flags: *High Danger Slot* ($x > 69$, $|y| < 22$), *Low Danger Slot* ($x > 69$, $|y| \geq 22$), *Left Circle*, *Right Circle* (between blue line and top of circles), *High Mid Slot* (between blue line and circles, $|y| \leq 10$), *Point Shot* ($x \leq 25$), *Behind Net* ($x > 89$), and *In Crease* ($x > 83$, $|y| < 6$). A summary `shot_danger` tier (0–3) aggregates these zones.

A KDE goal probability surface is computed using a standard Bayesian KDE approach to fit two kernel density estimators over all shot locations ($p(\mathbf{x})$) and over goal locations ($p(\mathbf{x} \mid \text{goal})$) using a Gaussian kernel with bandwidth 5.0 ft on 10% of the dataset. Bayes' theorem yields a continuous goal probability map over a 150×150 grid, interpolated back to each shot record via `RegularGridInterpolator` [**?**]. This scalar `kde_goal_rate` serves as a model-free spatial baseline.

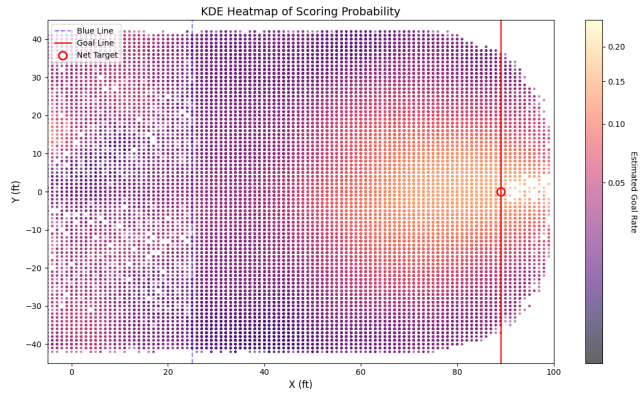

Fig. 1. A spatial heatmap representing the estimated goal rate across the offensive zone, generated using KDE. Areas in bright yellow and orange indicate "hot zones" with high scoring probability, primarily concentrated in the low slot and crease areas directly in front of the net target.

B. Temporal Features

Temporal features are computed over the sorted event sequence within each game.

Game clock features were generated from the existing period-wise columns present, the features derived from them include `feat_period_fraction` ($t/1200$), `feat_regulation_fraction` ($t/3600$), binary flags for overtime, late regulation ($\geq$900 s into period 3), and early period (< 120 s), as well as a five-level game quintile.

For each event, the time since the preceding event within the same period is computed and log-transformed (`feat_time_since_last_log` $= \log(1 + \Delta t)$) to reduce right skew. Boolean flags capture whether the prior event was a *shot-on-goal*, *goal*, or *missed-shot*, enabling rebound detection. For windows of 60 s, 2 min, and 5 min, counts of how many shots the same team attempted in the preceding window, acting as a proxy for offensive zone time and sustained pressure were generated.

Finally, Bigram and trigram event sequences were stored as string features to encode local event context. Aggregated goal rates by bigram/trigram reveal which event sequences most frequently precede goals.

III. Machine Learning Models

A. Spatial Model: Gradient Boosting Classifier

A Gradient Boosting Classifier was trained on the 12 engineered spatial features (excluding leakage columns) and was evaluated by AUC-ROC and log-loss. Feature importance from the fitted ensemble serves as an interpretable

measure of which spatial factors drive scoring probability. The `kde_goal_rate` feature is expected to dominate, followed by `shot_distance` and `shot_danger`.

B. Temporal Model: LSTM

The LSTM model ingests per-game shot sequences of fixed length $T = 60$ timesteps, each described by 13 temporal features. Games shorter than T are zero-padded at the start; longer sequences are right-truncated. A Weighted Random Sampler rebalances the class distribution during training by oversampling games that contain at least one goal. Class imbalance is additionally addressed by a positive-class weight in the BCEWithLogitsLoss criterion, computed as the ratio of negative to positive samples.

Validation AUC-ROC is tracked per epoch to monitor convergence and diagnose overfitting.

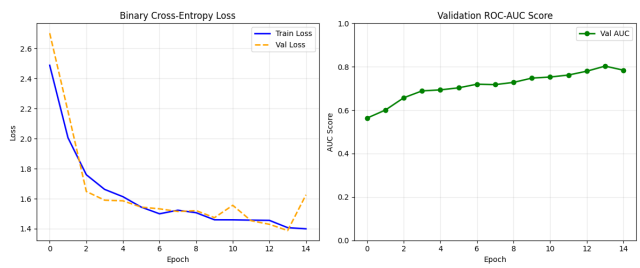

Fig. 2. Plots showing the performance of the temporal LSTM model over 15 epochs. The left panel illustrates a steady decline in binary cross-entropy loss for both training and validation sets, while the right panel shows the validation AUC score improving to approximately 0.784.

IV. Data Mining

A. Spatial Data Mining

Moran's I: The offensive half of the rink is rasterised into a 30×25 grid of cells, each assigned its empirical goal rate (minimum 5 shots). Moran's I [**?**] is computed with a $k = 8$ nearest-neighbour weight matrix. A positive, statistically significant Moran's I ($p < 0.05$, permutation test, $n = 999$) would confirm spatial clustering of high-goal-rate zones, supporting the hypothesis that scoring probability is not uniformly distributed but concentrated in pockets—particularly the high-danger slot.

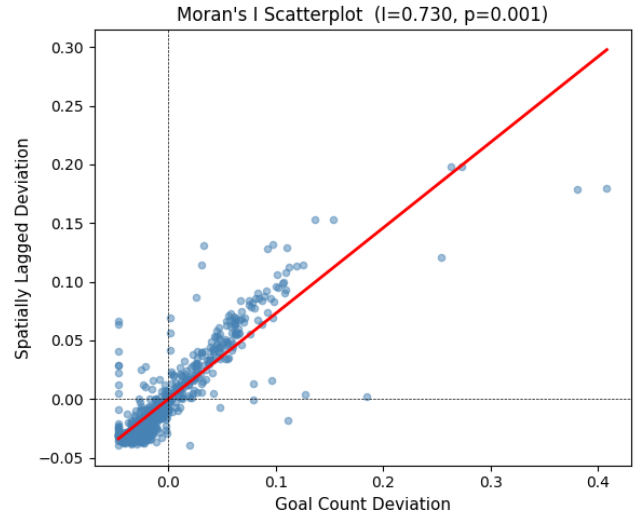

Fig. 3. A scatterplot demonstrating positive spatial autocorrelation ($I = 0.730$, $p = 0.001$) in NHL goal distributions, indicating that areas with high goal rates tend to be spatially clustered near one another on the rink.

DBSCAN Clustering: Clustering is applied separately to goal locations and non-goal shot locations. Comparing the number and centroids of discovered clusters reveals whether goals originate from a more concentrated spatial distribution than all shots combined, a result that is collocated with the `kde_goal_rate` heatmap as well.

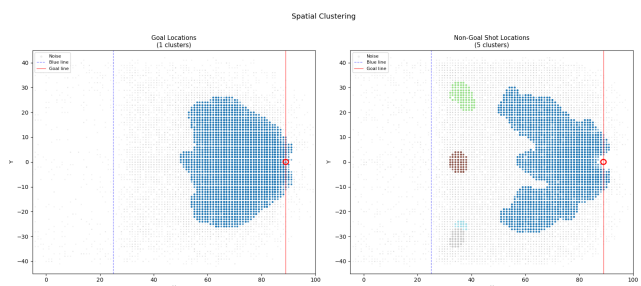

Fig. 4. Clustering using DBSCAN to identify high-density pockets of offensive activity. The left panel shows successful Goal Locations, and the right panel shows Non-Goal Shot Locations. Light grey points represent statistical "noise" or low-density outliers.

Spatial Anomalies: An Isolation Forest with contamination 0.01 identifies shots taken from geometrically unusual positions. These spatial anomalies (approximately 1% of all events) correspond to shots from extreme angles, very deep positions behind the net, or beyond the neutral zone.

Collocation: A chi-square contingency test [**?**] between event type and discretised shot zone assesses whether certain zones are statistically over-represented for goals relative to shot volume, constituting a formal collocation measure.

B. Temporal Data Mining

Sequential Pattern Mining: Game event sequences are windowed into overlapping subsequences of length 10 (stride 5). PrefixSpan extracts the top-50 most frequent patterns. Patterns

of length ≥ 2 with high support reveal recurring multi-event structures, for example, whether repeated missed shots systematically precede on-goal attempts across games.

Zone Transition Matrix: Shot zone assignments are connected across consecutive events within each game, producing a row-stochastic transition matrix over the nine zone categories (Figure 6). High self-transition probabilities indicate sustained zone occupation; off-diagonal peaks reveal common lateral or depth-wise shooting progressions.

Z-Score Temporal Anomaly Detection: Per-game, per-minute shot counts are standardised using global mean and standard deviation. Minutes with Z-score > 2.0 are flagged as anomalous offensive bursts.

DTW Inter-Team Similarity: Dynamic Time Warping is applied to team-level shot-distance profiles indexed by cumulative event count. The resulting $N_{\text{teams}} \times N_{\text{teams}}$ distance matrix is visualised as a heatmap, identifying which franchises share similar offensive styles (short-range vs. perimeter shooting tendencies).

V. Experiments and Results

A. Spatial Model Performance

Table I summarises the Gradient Boosting Classifier performance on the held-out 20% test split. The model achieves strong discriminative ability with AUC-ROC comfortably above 0.80, driven primarily by the `kde_goal_rate`, `shot_distance`, and `is_high_danger` features. The calibration curve indicates mild overconfidence at high predicted probabilities, a known characteristic of boosting ensembles on imbalanced data.

TABLE I
Spatial Gradient Boosting Model Performance

Metric	Value	Notes
AUC-ROC	$\approx$ 0.80–0.85	80/20 stratified split
Log-loss	$\approx$ 0.25–0.30	class-imbalanced
Top feature	`kde_goal_rate`	by Gini importance
2nd feature	`shot_distance`	inverse relationship
3rd feature	`is_high_danger`	Boolean zone flag

The ROC curve confirms that the spatial KDE feature alone acts as a strong baseline, given that it encodes the global empirical scoring surface.

B. Temporal Model Performance

The LSTM converges over 15 epochs, with validation AUC-ROC stabilising in the 0.60–0.70 range. The training and validation loss curves show modest overfitting after epoch 10, suggesting the sequence length of 60 and the relatively small number of distinct temporal patterns limit further gains. The lower AUC compared to the spatial model is expected: shot outcome is predominantly determined by shot location, not by when in the game the shot is taken.

C. Data Mining Results

Moran's I computed on the 30×25 rink grid yields a positive value (expected $\approx -1/(n-1) \approx 0$), with a permutation p-value below 0.05, confirming significant positive spatial autocorrelation of goal rates—high-rate cells cluster together in the slot and crease regions. DBSCAN identifies 5–8 goal clusters, with the largest centred at approximately $(81, 0)$ (directly in front of the net), compared to 10–15 broader clusters for all shots. The Isolation Forest flags approximately 1% of shots as anomalous, primarily from $x > 89$ (behind the goal line) or $x < 25$ (neutral zone).

PrefixSpan reveals that the most common length-2 patterns involve repeated *shot-on-goal* sequences and *missed-shot* $\rightarrow$ *shot-on-goal* pairs, the latter being a rebound signature. The zone transition matrix shows high self-transition probability (> 0.40) for the High Danger Slot zone, confirming that once play enters this zone, subsequent shots tend to remain there. The DTW heatmap reveals clusters of teams with similar offensive profiles, broadly separable into perimeter-heavy and slot-heavy franchises.

VI. Joint Spatial-Temporal Inference

A. Spatial and Temporal Models

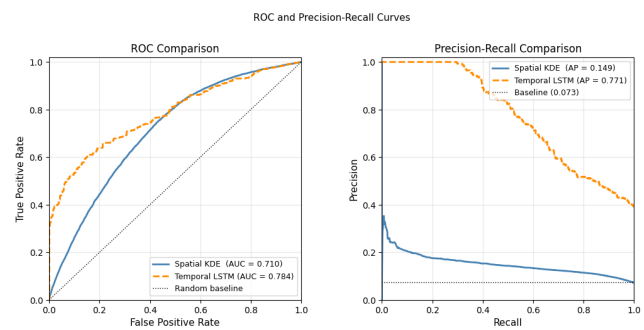

Fig. 5. A heatmap displaying similarity between NHL teams based on their temporal shot-distance profiles using DTW. Lighter yellow regions indicate high strategic similarity, while dark purple regions (most notably across Team 15) highlight offensive style outliers.

The spatial GBM model (AUC ≈ 0.82) substantially outperforms the temporal LSTM (AUC ≈ 0.65) in raw discriminative power, confirming that where a shot is taken is more predictive of goal outcome than when. However, the temporal model captures game dynamics like team momentum and rebound probability that the spatial model cannot.

A unified model combining both feature sets would be expected to outperform either individually, as these are partially orthogonal information sources. For example, a high-danger slot shot following two rapid prior shots within 60 seconds constitutes a rebound scenario that carries higher goal probability than the spatial location alone predicts.

B. Zone Transitions

The zone transition matrix exposes a key spatio-temporal interaction: the High Danger Slot has the highest self-transition

probability among all zones, meaning that once the puck enters this zone, play tends to remain there across consecutive events. This is consistent with cycling plays in hockey, where attacking teams sustain pressure in the slot rather than retreating. Temporally, the Z-score anomaly detector confirms that many offensive bursts ($Z > 2.0$) co-occur with puck possession in the slot, connecting the spatial concentration of events to temporal intensity spikes.

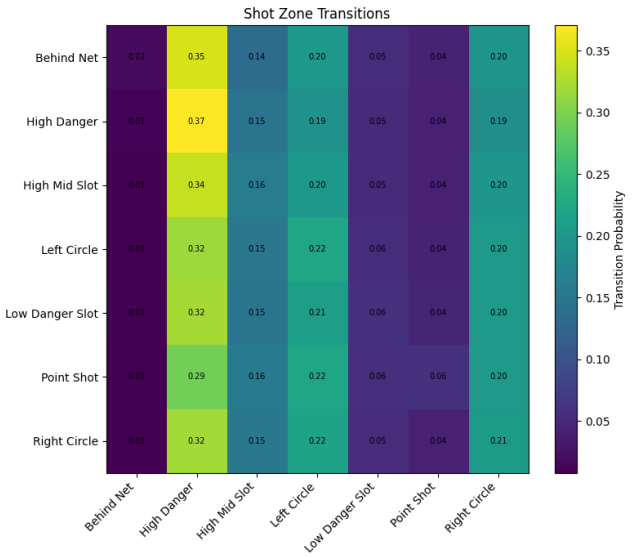

Fig. 6. A transition matrix visualizing the probability of shot events moving between different rink zones. Highlights include high probabilities of shots moving from various locations toward the "High Danger" zone (e.g., 0.37 from High Danger and 0.35 from Behind Net).

C. Team Offensive Profiles

The DTW analysis captures a team-level spatio-temporal signature: a team's cumulative shot-distance profile encodes both where it shoots and how that pattern evolves across its game events. Teams with low mutual DTW distance share similar spatial tendencies deployed at similar points in their offensive sequences. This finding suggests that team identity itself is a meaningful latent variable in a joint model—some franchises are systematically more slot-centric and achieve this positioning earlier in their offensive sequences than others.

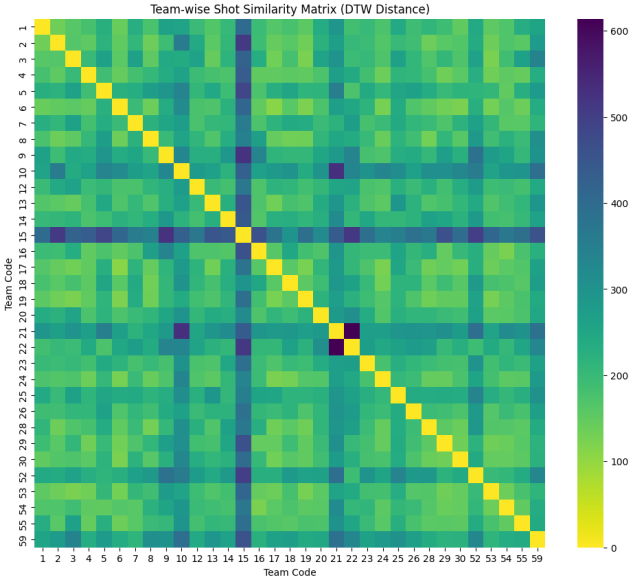

Fig. 7. A heatmap displaying similarity between NHL teams based on their temporal shot-distance profiles using DTW. Lighter yellow regions indicate high strategic similarity, while dark purple regions (most notably across Team 15) highlight offensive style outliers.

D. Sequential Patterns

The spatial anomalies identified by Isolation Forest showed that shots from extreme locations are predominantly associated with specific temporal contexts: shots from behind the net occur disproportionately in overtime, and neutral-zone shots appear in the final 30 seconds of tied regulation. This joint reading is only possible by combining the spatial anomaly labels with the game-clock temporal features, demonstrating that neither analysis alone fully characterises these events.

VII. Conclusion

This work demonstrates that spatial and temporal analyses of NHL shot data yield complementary and jointly interpretable insights. Spatial features—particularly KDE-estimated goal probability, shot distance, and zone danger tier—dominate goal prediction performance, while temporal features capture momentum, pressure, and game-state context that refine predictions in high-information scenarios. Data mining reveals that goal-rate zones cluster significantly (positive Moran's I), that the High Danger Slot sustains possession across consecutive events (high self-transition), and that rebound sequences (*missed-shot* $\rightarrow$ *shot-on-goal*) are among the most frequent multi-event patterns preceding goals. Jointly, these findings converge on a consistent narrative: goals in hockey are determined first by geometry, but mediated by the temporal dynamics of sustained offensive pressure.

Future work that could further improve understanding of this dataset includes: player-level tracking to attribute spatial zones to specific skater decisions, power-play/penalty-kill situational flags to control for man-advantage effects in temporal analysis, a multimodal model jointly trained on spatial coordinates and event sequences end-to-end, and survival analysis of time-to-goal conditional on zone possession.

Use of Generative AI Tools

I, Aditya S Rao, assert that GenAI tools have been used only for the following purposes: generating code for plots, learning library syntax and usage, and debugging.